

Trainium NKI | Kernel review & sanity check

One-page technical evidence brief | 10 Oct 2026 | Sources: combined.zip + final_review.pdf

Key result (reported): Level 4 tiled matmul reached reward 1.00 in 3/3 C2 runs using a memory plan plus memory-aware checker feedback. Baseline in the supplied PDF: 0/5 solves at reward 0.62. These are simulator results, not device benchmarks.

What the supplied code contains

Level / operation	Code-level observation	Reported verification
L1 / avg pool	HBM output; SBUF pooling windows; scalar output writes	4 seeds × 4 shapes
L2 / transpose	HBM output; Python per-element transpose assignments	4 seeds × 4 shapes
L3 / single matmul	SBUF return + self-copy DMA (hardware concern)	4 seeds × 1 shape
L4 / tiled matmul	128×128×512 output tiling; K tile 128; DMA + PSUM	4 seeds × 4 shapes

Why Level 4 matters

The kernel loops over M, N and K tiles, loads corresponding lhsT[k,m] and rhs[k,n] slices into SBUF, accumulates with nc_matmul in PSUM, then copies each output tile to shared HBM. The PDF attributes the final C2 fix to K-row provenance feedback that identified mismatched operand slices; it reports two distinct passing generated kernels.

Sanity checks performed here

PASS: ZIP CRC integrity; all eight bundled Python modules parse and compile. L4 indexing and output-tile coverage are structurally consistent when M, N and K are exact multiples of 128, 512 and 128 respectively.

NOT REPRODUCED: The 16/16 level-by-seed checks (52 individual shape/seed cases) are claims in README, not a local test result. The archive lacks nkibench.pristine.py, experiment logs, and an installed NKI runtime. The PDF also describes checker features, but their implementation is not in this ZIP.

Reviewer caveats & next validation

Hardware readiness: L3 returns an SBUF tile and copies it to itself rather than writing shared HBM; L1/L2 use scalar writes. L4 does not handle partial M/N/K tiles, and its PSUM initialization/accumulation semantics need device compilation confirmation. No Neuron hardware compilation, execution, latency, or bandwidth numbers were supplied.

Recommended next step: Bundle the pristine checker at its stated commit, fixed seed/shape logs, NKI version, and reproduce every case on the pod; then compile and run each kernel on Trainium before making hardware-performance claims.

Evidence note: “16/16” counts four seeds for each of four levels, not 16 independent shapes. The C2 3/3 vs baseline 0/5 contrast is exploratory (small sample; live checker changes disclosed in source PDF).